Fortinet Capstone Project

Enterprise Network Security Implementation

📌 Project Overview

Engineers are required to design, configure, and demonstrate a secure enterprise network using **FortiGate Firewalls** in a simulated lab environment (EVE-NG / GNS3 / PNET).

The project must simulate a real-world organization consisting of:

- Headquarters (HQ)
- Branch 1
- Branch 2
- Remote User

The objective is to implement secure connectivity, segmentation, high availability, VPN communication, routing, and security policies following enterprise best practices.

Note: IP addressing scheme is up to the student but must be logical and documented.

📦 Minimum Required Devices Per Site

The following numbers represent the **minimum required devices** to successfully complete all mandatory requirements.

🏢 Headquarters (HQ)

🔹 **Minimum Devices Required: 7 Devices**

1. **2 × FortiGate Firewalls**
   - Required for High Availability (Active–Passive)
2. **1 × ISP Router**
   - Simulates Internet provider
3. **1 × Edge/WAN Router**
   - Connects ISP to FortiGate cluster
4. **1 × Core Switch**
   - Handles VLAN trunking and LAN distribution
5. **2 × Access Switches**

- One for Users
- One for Servers / Management

🌐 **Branch 1**

🔹 **Minimum Devices Required: 4 Devices**

1. **1 × FortiGate Firewall**
2. **1 × ISP Router**
3. **1 × Edge/WAN Router**
4. **1 × Access Switch**

🌐 **Branch 2**

🔹 **Minimum Devices Required: 4 Devices**

1. **1 × FortiGate Firewall**
2. **1 × ISP Router**
3. **1 × Edge/WAN Router**
4. **1 × Access Switch**

💻 **Remote User**

🔹 **Minimum Devices Required: 1 Device**

1. **1 × Client Machine**
   - Used for SSL-VPN or IPsec Remote Access testing

ESSENTIAL REQUIREMENTS (Must-Have)

These requirements are mandatory. Missing any major section will significantly affect the grade.

1️⃣ Network Infrastructure

The lab must include:

- HQ site
- Branch 1 site
- Branch 2 site
- Remote user
- Proper LAN devices (switches/routers) inside sites
- ISP router simulation for WAN connectivity
- All devices must be correctly connected and operational

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2 HQ Configuration

VLAN Segmentation (Mandatory)

Create at HQ:

- User VLAN
- Server VLAN
- Management VLAN

Each VLAN must:

- Have its own subnet
- Be properly tagged
- Be connected via trunk to FortiGate

Inter-VLAN Routing

- Must be handled by the HQ FortiGate
- Verify communication between permitted VLANs
- Restrict access between sensitive VLANs where required

DHCP

- DHCP must be enabled for the User VLAN
- Proper IP addressing must be assigned dynamically

High Availability (HA) – Mandatory

At HQ:

- Two FortiGate devices
- Active–Passive configuration
- Heartbeat interfaces configured
- Configuration synchronization verified
- Failover test must be demonstrated

Failover test must show:

- Traffic continuity
- Role switching between primary and secondary

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3 Branch 1 & Branch 2 Configuration

Each branch must include:

VLANs

- User VLAN
- Server VLAN

Inter-VLAN Routing

- Must be handled locally by the branch FortiGate

Basic internal connectivity testing

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4 IPsec Site-to-Site VPN

Mandatory VPN tunnels:

- HQ ↔ Branch 1
- HQ ↔ Branch 2

Traffic requirements:

- Branch-to-HQ communication
- Branch-to-Branch communication (through HQ routing)

Proper phase 1 & phase 2 configuration required.

Static routes for VPN subnets must be configured.

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5 Firewall Policies

You must configure and demonstrate:

Internet Access Policies

- LAN → WAN
- With NAT enabled

VPN Policies

- LAN ↔ VPN
- Proper security inspection applied

Inter-VLAN Policies

- Allow only necessary access
- Restrict sensitive VLANs

Policies must follow least-privilege principle.

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6 NAT Configuration

Mandatory NAT settings:

- Outbound NAT for Internet access
- VIP (Virtual IP) for publishing at least one internal server

Demonstrate:

- External access to internal server
- Correct port forwarding

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7 Routing

You must configure:

- Static routes for all internal subnets
- Default route for Internet
- Routes for VPN networks
- Proper routing between sites

Routing must be clean and documented.

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8 Security Profiles (Mandatory)

Apply the following security profiles:

- Antivirus
- Web Filtering
- IPS
- Application Control

These profiles must be applied on:

- LAN → WAN policies
- VPN → LAN policies

Demonstrate blocking or detection behavior.

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9 Remote User VPN

Configure remote access VPN (SSL-VPN or IPsec client).

Demonstrate:

- Successful connection
- Access to internal resources
- Proper user authentication

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10 Documentation + Submission Requirements

Students must submit:

1. Full Documentation (PDF)

Including:

- IP addressing table
- VLAN table
- Routing table
- VPN configuration summary
- Security policy explanation
- NAT explanation
- HA explanation
- Test results screenshots

2. Presentation (PowerPoint)

Explaining:

- Architecture overview
- Security design
- Key features implemented
- Challenges faced

3. Demo Video

Must demonstrate:

- HA failover
- VPN connectivity
- Security profile blocking
- NAT functionality
- Internet access
- Branch-to-Branch communication

⭐ BONUS REQUIREMENTS (Extra Credit)

Implementing any of the following gives additional marks:

🔹 SD-WAN Configuration

- Multiple ISP links
- Traffic steering
- SLA monitoring

🔹 DNS Filtering

Enable DNS security profile and demonstrate domain blocking.

🔹 DoS Policies

Configure DoS protection and simulate detection.

🔹 Certificates / PKI

- Import or generate certificates
- Use certificate-based authentication
- Enable HTTPS inspection

🔹 Security Fabric Integration

- Enable Security Fabric
- Show device visibility
- Centralized monitoring

🔹 Logging Server

- Configure FortiAnalyzer OR
- Configure Syslog server
- Demonstrate log collection

🔹 Automation Stitches

Create automation such as:

- Alert when VPN goes down
- Alert when HA failover occurs
- Email notification on IPS trigger

🔹 Advanced Application Control

- Block specific apps (e.g., Torrent, VPN apps)
- Use category-based filtering

🔹 Server Hardening

Demonstrate:

- Restricted access
- Limited ports open
- Controlled exposure through VIP

CREATIVITY BONUS

Students will receive extra credit for:

- Professional enterprise design
- Clean segmentation strategy
- Advanced routing (OSPF / BGP)
- Zero Trust approach
- Micro-segmentation
- Security best-practice alignment
- Clear logical structure
- Realistic IP planning
- Clean policy organization
- Professional presentation quality

Any additional security solution beyond requirements is considered bonus.

Examples:

- NAC simulation
- MFA integration
- Threat intelligence feeds
- Geo-blocking
- Sandbox simulation
- Advanced logging dashboards

Final Objective

By the end of this project, students must demonstrate:

- Enterprise network design
- Secure segmentation
- High availability
- Secure site-to-site connectivity
- Threat inspection
- Proper routing and NAT
- Professional documentation skills